3. Score based perspective: From EBMs to NCSNs

3W3 Mon 3/16 ~ 3W4 Mon 3/23

Monday, March 16, 2026 8:10 PM

3.1. Energy based Models 3W3 Mon 3/16

$p_\phi(x) := \frac{\exp(-E_\phi(x))}{Z_\phi}$ $\quad Z_\phi := \int_{\mathbb{R}^D} \exp(-E_\phi(x))\,dx$.

Low Energy → High Prob.

⇒ Z_ϕ?? Intractable.

Score

For density $p(x): \mathbb{R}^D \to \mathbb{R}$, score function is the gradient of log likelihood.

$S(x) = \nabla_x \log p(x) \ : \mathbb{R}^D \to \mathbb{R}^D$

Advantages of score function estimations.

1. Freedom of partition function.

$\nabla_x \log p(x) = \nabla_x[-E(x) - \log Z] = -\nabla_x E(x)$.

⇒ Score function은 좋다.

2. Complete Representation.

미적분학의 기본정리에 의해 자명.

Training EBMs w/ Score functions. (Score Matching)

$\mathcal{L}_{SM}(\phi) = \frac{1}{2}\mathbb{E}_{x_0 \sim p_{data}(x_0)} \| \nabla_x \log p_\phi(x) - \nabla_x \log p_{data}(x_0) \|_2^2$

⇒ 전개하면 자명히 $\nabla_x \log p_{data}(x_0)$와 Independent, 아주 쉬움.

Sampling from EBMs.

We know $E_\phi(x)$ ⇒ How to sample?

Sampling ⇔ Simulating a particle in this Boltzmann system.

$p(x) \propto e^{-\frac{1}{k_B T}E_\phi(x)}$

$m\frac{d^2}{dt^2}X = \underbrace{-\nabla_x E_\phi(x)}_{=\text{Force}} - \underbrace{\gamma \frac{d}{dt}X}_{=\text{Damping}} = \underbrace{0}_{=\text{Assumption}}$. 왜 가속도가 0으로 가정?

⇒ Deterministic ODE : $dX = -\frac{1}{\gamma}\nabla_x E_\phi(x)\,dt$.

But w/ this ODE, all particles end up in a minima.

We want the density function to be constant!

∴ SDE : $dX = -\frac{1}{\gamma}\nabla_x E_\phi(x)\,dt + \alpha\, dw_t$ $\quad$ Brownian Motion $W(t+\Delta t) - W(t) \sim N(0, \Delta t I)$

$W_t \sim N(0, dt \cdot I)$

Fokker-Plank Equation!

$dX = A(x)\,dt + B(x)\,dw_t \Rightarrow \frac{\partial}{\partial t}\rho = -\nabla \cdot (\rho A) + \frac{1}{2}\nabla^2(\rho B^2)$

PROOF?

∴ Constant PDF : $\frac{\partial}{\partial t}\rho = -\nabla \cdot \left(-\frac{1}{\gamma}\rho \nabla E_\phi(x)\right) + \frac{1}{2}\nabla^2(\rho \alpha^2)$ $\quad \rho \sim e^{-\frac{E}{kT}}$, $\nabla\rho = -\frac{\nabla E}{kT}\rho$

$= -\frac{kT}{\gamma}\nabla \cdot (\nabla\rho) + \frac{1}{2}\alpha^2 \nabla^2 \rho$

$\alpha^2 = \frac{2kT}{\gamma}$

∴ Langevian Dynamics SDE update : $\underline{dX = -\frac{1}{\gamma}\nabla E \cdot dt + \sqrt{\frac{2kT}{\gamma}}\,dw_t}$

⇒ Euler-Maruyama Discretization : $X_{n+1} - X_n = -\eta \nabla E(x_n) + \sqrt{2\eta}\,\varepsilon_n$ $\quad kT = 1$, $\varepsilon_n \sim N(0, I)$

Cons : Mixing Time, Sensitivity per η, Struggles to transit between modes.

3.2. From Energy Based to Score Based Generative Models. 3W3 Thu 3/19

Score Matching

We showed that matching scores can be a variation, with sampling through Langevin Dynamics.

$S_\phi(x) \simeq S_{data}(x)$ $\quad S(\cdot) := \nabla_x \log p(\cdot)$

$\mathcal{L}_{SM}(\phi) = \frac{1}{2}\mathbb{E}_{x \sim p_{data}} \| S_\phi(x) - S_{data}(x) \|_2^2$

But S_{data} is unknown.

Hyvärinen's tractable form

$\mathcal{L}_{SM}(\phi) = \mathbb{E}_{x \sim p_{data}}\left[\mathrm{Tr}(\nabla_x S_\phi(x)) + \frac{1}{2}\|S_\phi(x)\|_2^2 \right] + C$.

proof.

$\mathcal{L}_{SM}(\phi) = \frac{1}{2}\mathbb{E}_{x \sim p_{data}}(S_\phi(x) - S_{data}(x)) \cdot (S_\phi(x) - S_{data}(x))$

$= \frac{1}{2}\mathbb{E}_{x \sim p_{data}}\left[\|S_\phi(x)\|_2^2 + \|S_{data}(x)\|_2^2 - 2 S_\phi(x) \cdot S_{data}(x) \right]$

$= \frac{1}{2}\mathbb{E}_{x \sim p_{data}}\|S_\phi(x)\|_2^2 + C + \frac{1}{2}\int_{\mathbb{R}^D} -2 S_\phi(x) \cdot \nabla_x \log p_{data}(x) \cdot p_{data}(x)\,dx$ $\quad \left(\nabla_x \log p \cdot p = \frac{\nabla_x p}{p} \cdot p = \nabla_x p\right)$.

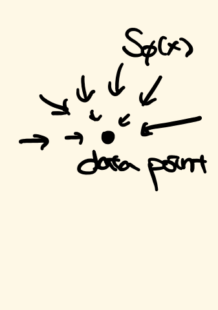
$= \frac{1}{2}\mathbb{E}_{x \sim p_{data}}\|S_\phi(x)\|_2^2 + C - \left[\underline{\left[S_\phi(x) \cdot p_{data}(x) \mathbb{I} \right]_{\partial \mathbb{R}^D}} - \int_{\mathbb{R}^D} \nabla_x S_\phi(x) \cdot p_{data} \mathbb{I}\,dx \right]$ (Integral by Parts)

$= 0$ for $x \to \infty \Rightarrow S \cdot p \to 0$. Stein's Identity.

$= \mathbb{E}_{x \sim p_{data}}\left[\mathrm{Tr}(\nabla_x S_\phi(x)) + \frac{1}{2}\|S_\phi(x)\|_2^2 \right] + C$ ∎

각 항의 의미 : $\|S_\phi(x)\|_2^2$: data sample 에서 작음.

$\mathrm{Tr}(\nabla_x S_\phi(x))$: data sample 에서 감소함.



⇒ Score becomes stationary around high density areas.

cf. Sampling w/ Langevin Dynamics.

$\nabla E(x_n) \to S_\phi(x_n)$

3.3. Denoising Score Matching 3W3 Fri 3/20

Trace of the jacobian $\mathrm{Tr}(\nabla_x S_\phi(x))$ is $O(D^2)$ in computation.

$S_\phi(x) = \begin{bmatrix} S_\phi^{(1)} \\ \vdots \\ S_\phi^{(D)} \end{bmatrix}$ $\quad \|S_\phi(x)\|_2^2 = \sum_{k=1}^{D} S_\phi^{(k)}(x)^2$

$\nabla_x S_\phi(x) = \begin{bmatrix} \partial/\partial x_1 \\ \vdots \\ \partial/\partial x_D \end{bmatrix} [S_\phi^{(1)} \cdots S_\phi^{(D)}]$ $\quad \mathrm{Tr}(\nabla_x S_\phi(x)) = \sum_{k=1}^{D} \frac{\partial}{\partial x_k} S_\phi^{(k)}(x)$

Slicing Score Matching (Computing for random directions).

Let. $u \in \mathbb{R}^D$ (isotropic vector). $\mathbb{E}[u] = 0$ $\quad \mathbb{E}[uu^T] = I$.

Then, $\mathbb{E}_u[u^T A u] = \underbrace{\mathbb{E}_u[u_i A_{ij} u_j]}_{\text{(Einstein's Notation)}} = \sum_{i,j} \delta_{ij} A_{ij} = \sum_k A_{kk} = \mathrm{Tr}(A)$ $\quad \mathbb{E}[u_i u_j] = \delta_{ij}$.

$\mathbb{E}_u[(u^T S_\phi(x))^2] = \mathbb{E}_u[(\sum_i u_i S_\phi^{(i)}(x))^2] = \mathbb{E}_u[\sum_{i,j} u_i u_j S_\phi^{(i)}(x) S_\phi^{(j)}(x)]$

$= \sum_{i,j} \delta_{ij} S_\phi^{(i)}(x) S_\phi^{(j)}(x) = \sum_k S_\phi^{(k)}(x)^2 = \|S_\phi(x)\|_2^2$.

∴ $\mathcal{L}_{SM}(\phi) = \mathbb{E}_x\left[\mathrm{Tr}(\nabla_x S_\phi(x)) + \frac{1}{2}\|S_\phi(x)\|_2^2 \right]$

$= \mathbb{E}_{u,x}\left[\underline{u^T \cdot \nabla_x S_\phi(x) \cdot u} + \frac{1}{2}(u^T S_\phi(x))^2 \right]$.

Use JVP for efficiency.

Trace doesn't need explicit Jacobian as well?

Denoising Score Matching

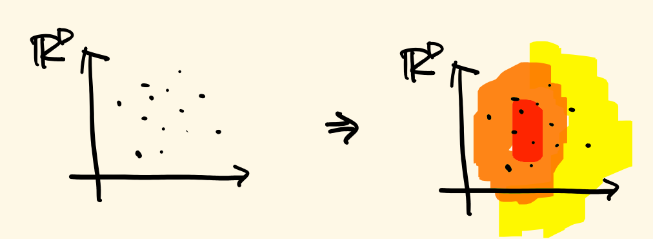
Computation, Normalization, Sampling, Training → Solved.

But this method has a high data dependency.

S_{data} is intractable because we have a discrete data sample set.

⇒ If we define p_{data} as a sum of gaussians around data points,

p_{data} becomes tractable + differentiable!



Why not sample directly from p_{data}? Too much computation?

⇒ A conditional pdf. $p_\sigma(\tilde{x}|x) = N(\tilde{x}; x, \sigma^2 I)$.

Our differentiable p_{data} : $p_\sigma(\tilde{x}) = \int_{\mathbb{R}^D} p_\sigma(\tilde{x}|x)\, p_{data}(x)\,dx$. (Marginal).

Here, our conditional distribution is differentiable, but marginal is intractable.

$\mathcal{L}_{SM}(\phi;\sigma) = \frac{1}{2}\mathbb{E}_{\tilde{x} \sim p_\sigma(\tilde{x})}\left[\|S_\phi(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\|_2^2 \right]$ $\quad$ Marginal → Conditional.

$= \frac{1}{2}\mathbb{E}_{x \sim p_{data}(x),\, \tilde{x} \sim p_\sigma(\tilde{x}|x)}\left[\|S_\phi(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|x)\|_2^2 \right] + C$

Since $\mathcal{L}_{DSM}$ uses a predefined conditional distribution, we have a closed form for a loss of better coverage.

$\mathcal{L}_{DSM}(\phi;\sigma) = \frac{1}{2}\mathbb{E}_{x \sim p_{data}(x),\, \tilde{x} \sim p_\sigma(\tilde{x}|x)}\left[\left\| S_\phi(\tilde{x}) - \nabla_{\tilde{x}} \log \frac{e^{-\frac{1}{2\sigma^2}\|\tilde{x}-x\|_2^2}}{Z} \right\|_2^2 \right]$

$= \frac{1}{2}\mathbb{E}_{x \sim p_{data}(x),\, \tilde{x} \sim p_\sigma(\tilde{x}|x)}\left[\left\| S_\phi(\tilde{x}) + \frac{2}{2\sigma^2}\|\tilde{x} - x\| \right\|_2^2 \right]$

$= \frac{1}{2}\mathbb{E}_{x \sim p_{data}(x),\, \varepsilon \sim N(0,I)}\left[\left\| S_\phi(x + \sigma\varepsilon) + \frac{\varepsilon}{\sigma} \right\|_2^2 \right]$

cf. Sampling happens w/ Langevin Dynamics.

Why DSM is Denoising (Score = Noise).

Tweedie's Formula. : States that a denoised estimate x from a corrupt gaussian pdf $p_\sigma(\tilde{x})$ is given via taking σ^2 step in the direction of the score $\nabla_{\tilde{x}} p_\sigma(\tilde{x})$

⇒ Let. $p_\sigma(\tilde{x}|x) = N(\tilde{x} | \alpha x, \sigma^2 I)$ $\quad x \sim p_{data}(x)$.

Then, $\alpha \cdot \mathbb{E}_{x \sim p(x|\tilde{x})}[x] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$.

PROOF.

$p_\sigma(\tilde{x}) = \int_{\mathbb{R}^D} p_\sigma(\tilde{x}|x)\, p_{data}(x)\,dx$

$\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}) = \int_{\mathbb{R}^D} \frac{p_{data}(x)}{p_\sigma(\tilde{x})} \cdot \nabla_{\tilde{x}} p_\sigma(\tilde{x}|x)\,dx$.

$\nabla_{\tilde{x}} p_\sigma(\tilde{x}) = \int_{\mathbb{R}^D} \nabla_{\tilde{x}}\left[\frac{1}{Z} e^{-\frac{1}{2\sigma^2}\|\tilde{x} - \alpha x\|_2^2} \right] p_{data}(x)\,dx$.

$= \int_{\mathbb{R}^D} p_\sigma(\tilde{x}|x) \cdot \left(-\frac{1}{\sigma^2}(\tilde{x} - \alpha x)\right) p_{data}(x)\,dx$.

$= -\frac{1}{\sigma^2}\left(\tilde{x} - \alpha \mathbb{E}_{x \sim p(x|\tilde{x})}[x]\right) \cdot p_\sigma(\tilde{x})$ $\quad$ Bayes.

∴ $\alpha \mathbb{E}_{x \sim p(x|\tilde{x})}[x] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$.

Tweedie's Formula ≃ Bayes's Rule. Noise sample → clean sample Estimation.

in DSM, $\alpha = 1$. $\quad S_\phi^*(\tilde{x};\sigma) = \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$.

* $\mathcal{L}_{DSM}$ 생긴거 보면 자명??

SURE (Stein's Unbiased Risk Estimator)

We now know that Score Matching = Denoising.

Let. denoiser $D(\tilde{x}) : \mathbb{R}^D \to \mathbb{R}^D$

Then, $\mathcal{L}_{DSM} = \mathbb{E}_{x \sim p_{data},\, \varepsilon \sim N}\left[\|D(x + \sigma\varepsilon) - x\|_2^2 \right]$

⇒ Depends on clean sample! 이게 왜 문제? New Gaussian sample per data point?

cf. Stein's Identity. (Integral by Parts).

$\mathbb{E}_{z \sim \phi(z)}[\nabla_z g(z)] = \mathbb{E}_{z \sim \phi(z)}[z g(z)]$ $\quad g: \mathbb{R}^D \to \mathbb{R}$ $\quad z \in \mathbb{R}^D$

$\mathbb{E}_{z \sim \phi(z)}[\nabla_z \cdot u(z)] = \mathbb{E}_{z \sim \phi(z)}[z^T \cdot u(z)]$ $\quad u: \mathbb{R}^D \to \mathbb{R}^D$.

Using Stein's Identity,

$\mathcal{L}_{DSM}(D, x) = \mathbb{E}_{\varepsilon \sim N(0,I)}\left[\|D(\tilde{x}) - \tilde{x}\|_2^2 + 2(\tilde{x} - x)^T(D(\tilde{x}) - x) + \|\tilde{x} - x\|_2^2 \right]$ $\quad = \|D(\tilde{x}) - \tilde{x} + \tilde{x} - x\|_2^2$

$= \mathbb{E}_{\varepsilon \sim N(0,I)}\left[\|D(\tilde{x}) - \tilde{x}\|_2^2 + 2\sigma^2 \nabla_{\tilde{x}} D(\tilde{x}) - \sigma^2 D \right]$

(Stein's Identity).

각 항의 의미. $\|D(\tilde{x}) - \tilde{x}\|_2^2 - \sigma^2 D$: D이 크는 것이 이상적.

하지만, $D(\tilde{x}) = \tilde{x}$ 인 경우가 최소 $(-\sigma^2 D)$

$\nabla_{\tilde{x}} D(\tilde{x})$: 잘 학습된 Denoiser. $\nabla_{\tilde{x}} D(\tilde{x}) = \nabla_{\tilde{x}} x = 0$.

if. $D(\tilde{x}) = \tilde{x}$, $\nabla_{\tilde{x}} D(\tilde{x}) = \nabla_{\tilde{x}} \tilde{x} = D$.

∴ $D(\tilde{x})$가 상수함수이면서 $\|D(\tilde{x}) - \tilde{x}\|_2^2 - \sigma^2 D$ 을 최소화 = Denoiser.

3.4. Multi Noise Levels of DSM (NCSN) 3W4 Mon 3/23

Single Layer σ is unstable

⇒ Big σ → Small σ in sampling, Weighted sum loss for all Noise Levels.

Once again, Denoising = Score Matching (DDPM = NCSN).

$S^*(x_\sigma, x) = -\frac{1}{\sigma}\varepsilon^*(x_\sigma, \sigma)$.

⇒ DDPM + NCSN → SDE Framework.